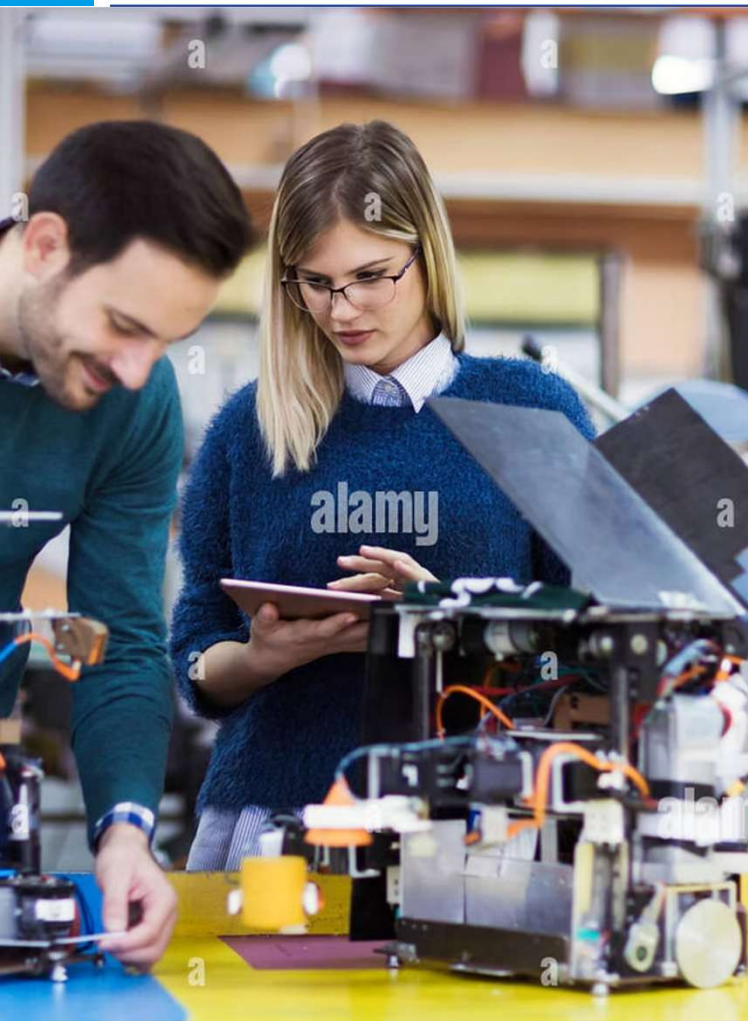




Robotics & Computer Vision

Week 1 — History, Classification & Applications



Course Overview



Combination

Robotics + Computer Vision — two fields united in one powerful curriculum



14 Weeks

From foundational theory to hands-on hardware integration



Language

Python + Robotics Toolbox by Peter Corke



Goal

Build AI-powered robots with integrated computer vision capabilities

PART 1

History of Robotics

From early mechanical automata to intelligent autonomous systems — tracing the remarkable journey of robotics across eight decades of innovation.



What Is a Robot?

Core Definition

A robot is an automatic machine controlled by a computer program — designed to perceive its environment and take actions to achieve specific goals.

Three Essential Components

→ Actuators

Mechanical mechanisms that physically move or manipulate objects

→ Sensors

Devices that perceive the environment (cameras, encoders, IMUs)

→ Processor / Controller

The computational brain that processes data and generates commands

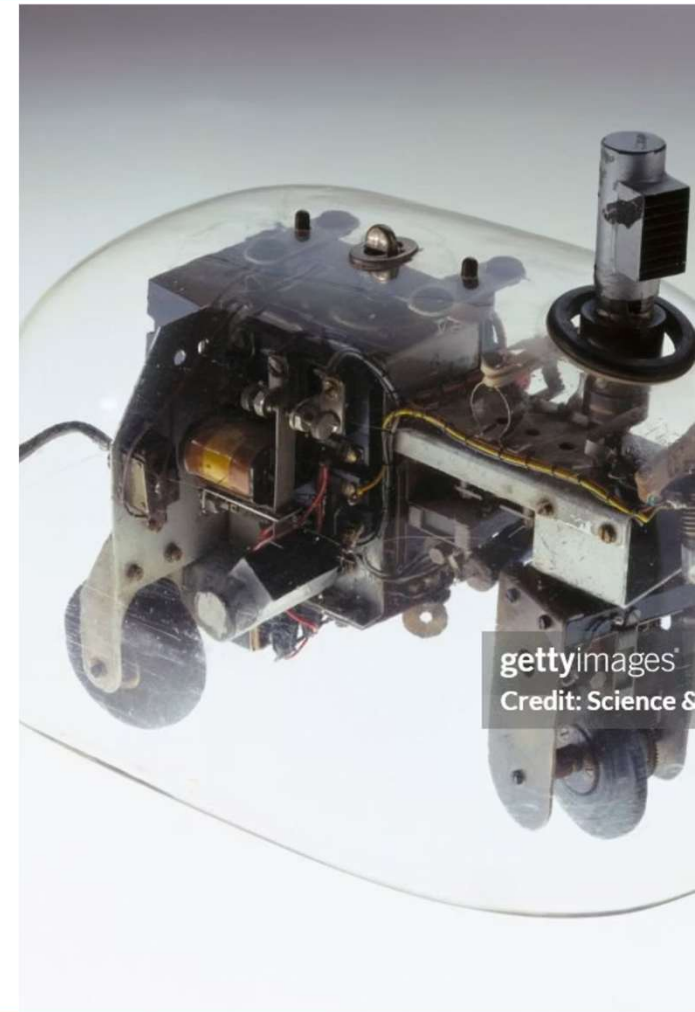
The Early Era (1940s–1950s)

1948 — Robotic Tortoise

William Grey Walter created the first autonomous robotic turtles, called **Elmer and Elsie**. These small machines could react to light stimuli and navigate around obstacles, demonstrating that simple neural circuits could produce surprisingly lifelike behavior.

1954 — Unimate

George Devol invented the world's first programmable robotic arm — **Unimate**. This landmark invention laid the foundation for all modern industrial robots and earned Devol recognition as the "Father of Robotics."



The Industrial Revolution (1960s)

The 1960s marked the dawn of industrial robotics as machines left the laboratory and entered the factory floor.

1961: Unimate at General Motors

The first Unimate robot was installed at a GM plant in New Jersey, performing dangerous spot-welding tasks that humans found hazardous and exhausting.

Impact on Manufacturing

Robots dramatically increased throughput and precision, performing repetitive and dangerous operations 24/7 without fatigue — setting a new standard for industrial productivity.



The Industrial Revolution (1960s)

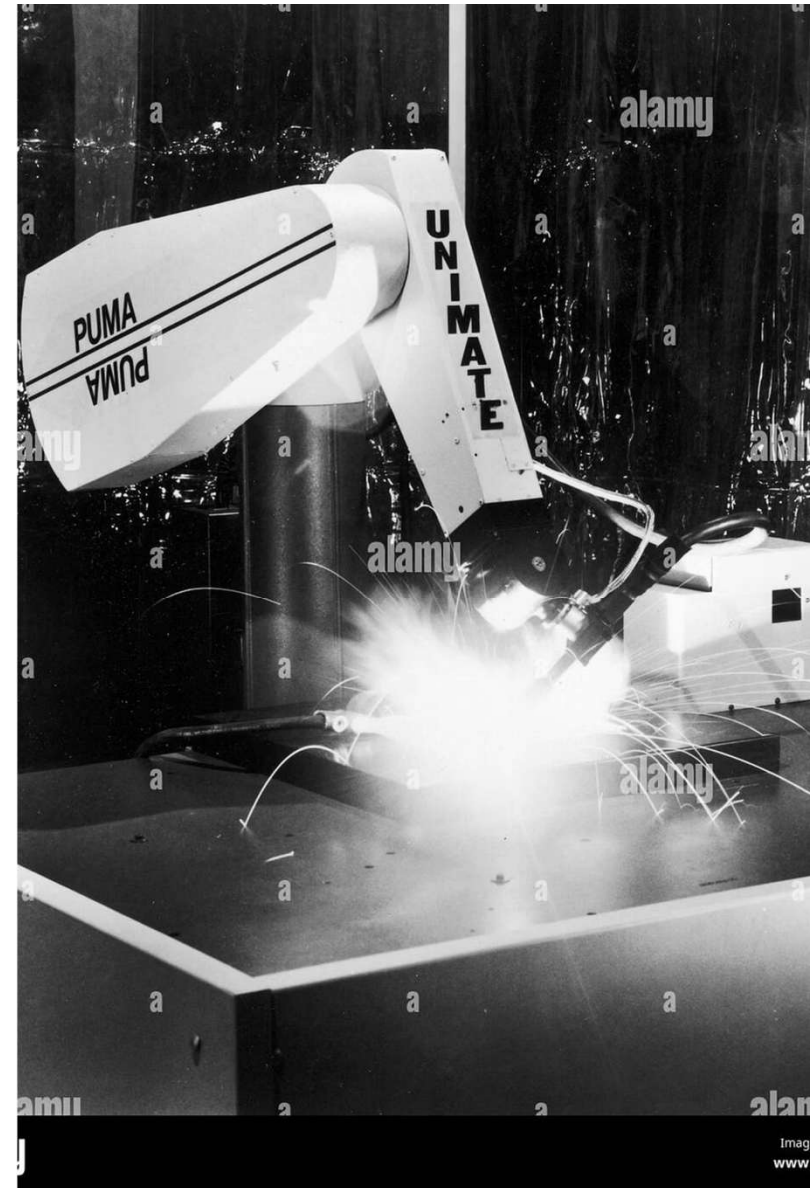
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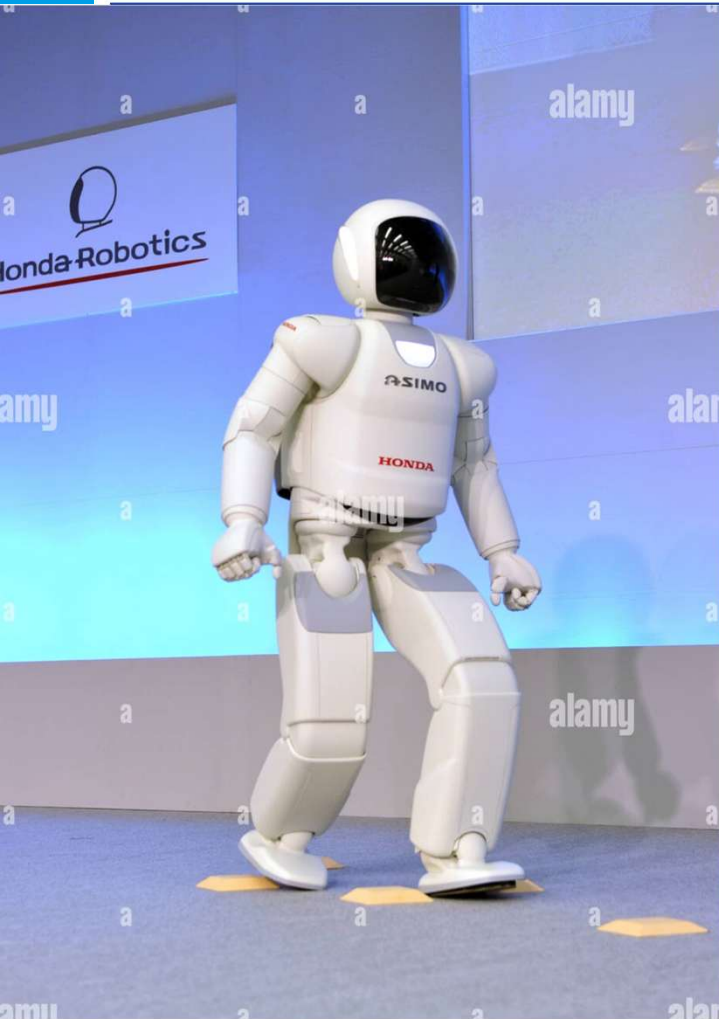
The Development Era (1970s–1980s)

1969 — Stanford Arm

Developed at Stanford University, this **6-DOF computer-controlled robotic arm** became the standard reference design for robot manipulators. It introduced the concept of software-controlled motion planning that remains fundamental today.

1970s — WABOT Project (Japan)

Japan's Waseda University launched the **WABOT project**, developing the world's first anthropomorphic robot capable of limited conversation and object manipulation. This pioneered the field of humanoid robotics and placed Japan at the forefront of the field.



The Modern Era (1990s–2010s)

- 1** — **2000**
Honda **ASIMO** — the first humanoid robot to walk bipedally with dynamic balance, climb stairs, and interact with humans
- 2** — **2002**
iRobot **Roomba** — the first commercially successful autonomous vacuum robot, bringing robotics into everyday households
- 3** — **2010s**
Collaborative Robots (Cobots) — robots designed to safely work alongside humans on shared tasks

Recent Breakthroughs (2015–2024)



AI + Deep Learning

Neural networks now run directly on-board robots for real-time perception and decision-making



Multi-Sensor Fusion

Combining camera, LiDAR, and IMU data for richer environmental understanding



SLAM

Simultaneous Localization and Mapping enables robots to explore unknown environments autonomously

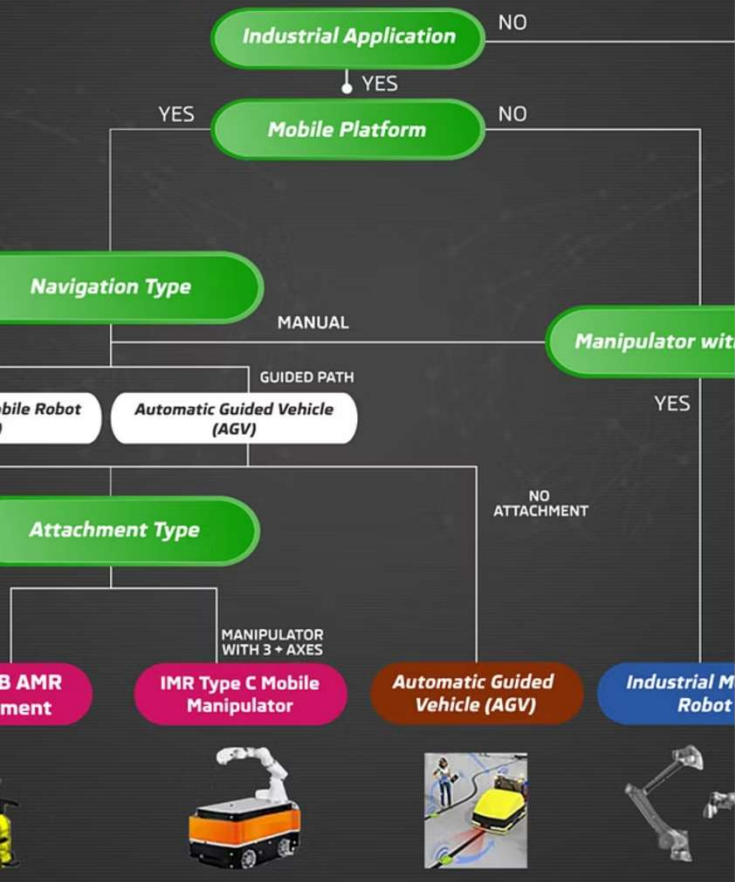


New Humanoids

Boston Dynamics, Tesla Optimus, and others pushing humanoid robots into real-world deployments



Classification of Industrial Robots - ANSI/RIA



PART 2

Robot Classification

Robots come in an enormous variety of forms and capabilities. Understanding how they are classified — by morphology, mobility, and function — helps us choose the right platform for the right task.

Classification by Form Factor



Robotic Arms (Manipulators)

Fixed-base or platform-mounted arms that perform precise manipulation tasks. They excel in manufacturing, welding, painting, and assembly where repeatability is critical.



Mobile Robots

Robots that move through their environment using wheels, legs, or wings. They are used for exploration, delivery, inspection, and navigation tasks where reach matters.

Types of Robot Manipulators

1

Articulated

Multiple rotary joints mimicking a human arm. Most common in industry. Highly flexible workspace, ideal for welding, painting, and general assembly.

2

SCARA

Selective Compliance Arm for Robotic Assembly. Rigid in the vertical axis but compliant horizontally — perfect for high-speed pick-and-place tasks.

3

Cartesian (Gantry)

Moves along orthogonal X-Y-Z axes. Simple, precise, and highly repeatable. Common in CNC machines and 3D printers.

4

Delta (Parallel)

Three arms connected to a common end-effector. Extremely fast and light — used in high-speed food packaging and pharmaceutical pick-and-place.

Types of Mobile Robots



Wheeled

2 to 4+ wheels — fast, energy-efficient, best for flat structured environments like warehouses and offices



Tracked

Tank-like tracks for rough terrain, rubble, and stairs — used in military and disaster response scenarios



Legged

Bipedal humanoids or quadrupeds — navigate stairs, uneven ground, and human environments naturally



Flying (UAV)

Unmanned aerial vehicles — quadcopters and fixed-wing drones for surveillance, mapping, and delivery



Classification by Function

Industrial

Manufacturing, welding, assembly, and material handling in controlled factory environments

Service

Customer service, cleaning (Roomba), logistics, hotel concierge, and public-facing tasks

Medical

Surgical assistance (da Vinci), rehabilitation exoskeletons, and drug dispensing systems

Research

Experimental platforms for advancing AI, locomotion algorithms, and human-robot interaction

Robot Generations

1

Gen 1 (1950–1970)

Fixed logic, no sensory feedback — pure mechanical automation

2

Gen 2 (1970–1990)

Programmable behavior, basic sensors for force and position feedback

3

Gen 3 (1990–2010)

Adaptive programming, rich environment interaction, computer vision integration

4

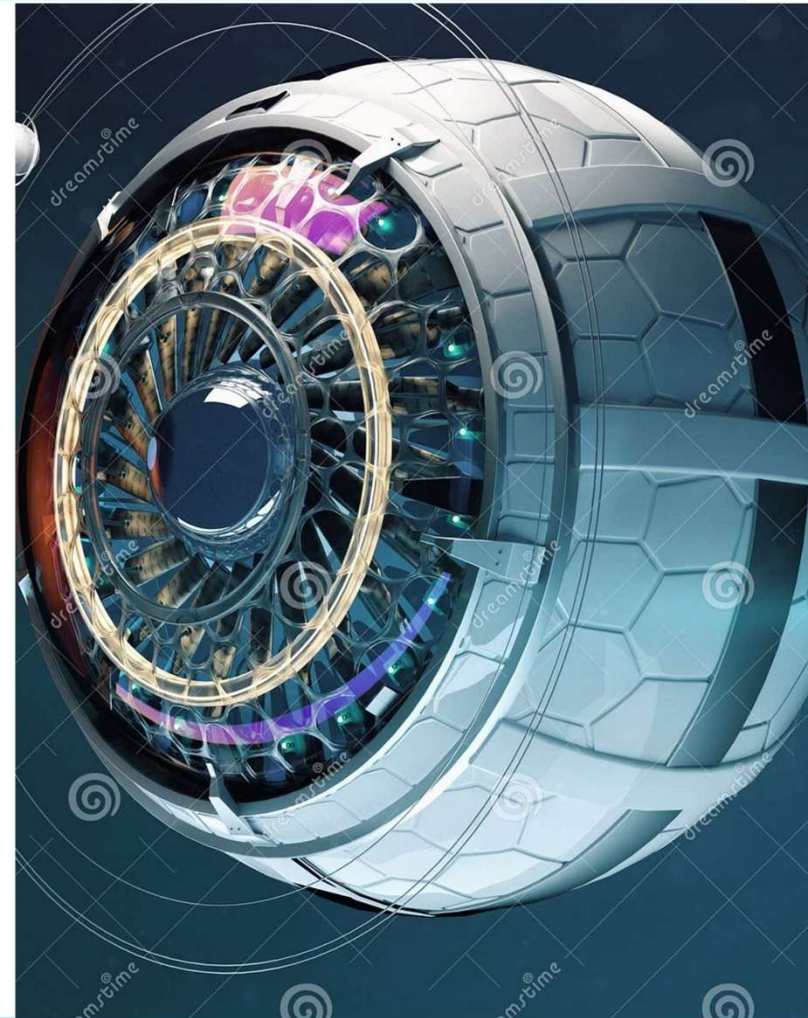
Gen 4 (2010–Present)

AI-driven learning, multi-robot collaboration, and autonomous decision-making

PART 3

History of Computer Vision

The quest to give machines the gift of sight spans over six decades — from simple edge detectors to deep neural networks that outperform human visual perception.



What Is Computer Vision?

Core Concept

Computer Vision is the scientific discipline that enables machines to **extract meaning from images and video**, replicating the human visual system's ability to interpret the world.

Key Capabilities

→ Image Processing

Filtering, enhancing, and transforming raw pixel data

→ Object Detection

Locating and classifying objects within a scene

→ Recognition

Identifying faces, text, gestures, and semantic content



Early Beginnings (1950s–1960s)

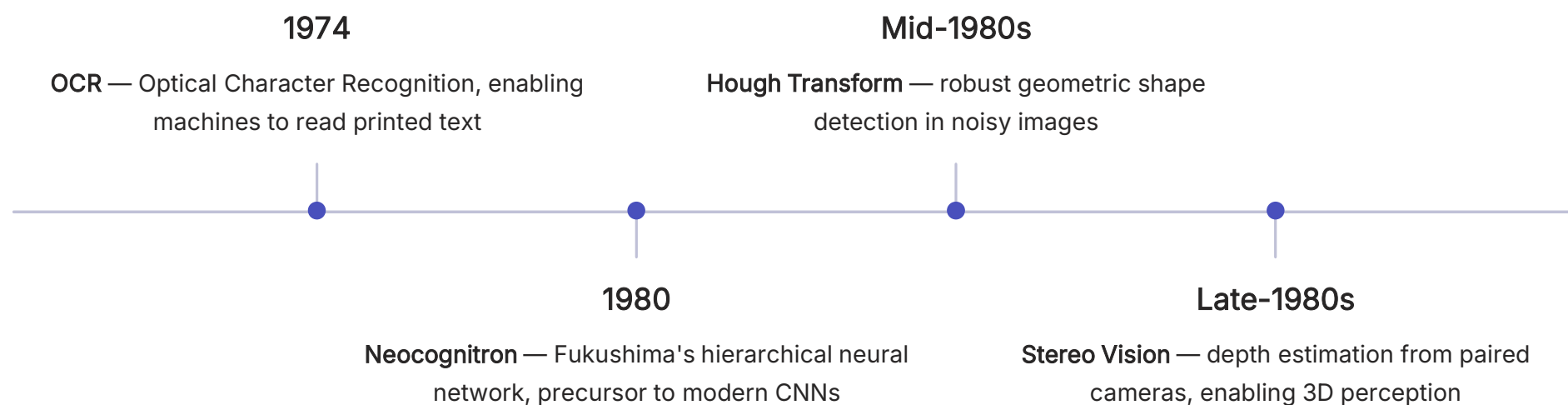
1950s — The Perceptron

Frank Rosenblatt introduced the **Perceptron** — the first artificial neural network capable of learning from examples. It was designed to classify visual patterns, making it the conceptual ancestor of all modern deep learning.

1963 — 3D Scene Understanding

Lawrence Roberts demonstrated that a computer could **interpret 3D structure from 2D images** by analyzing edges and geometric relationships — a landmark achievement that opened the entire field of computational vision.

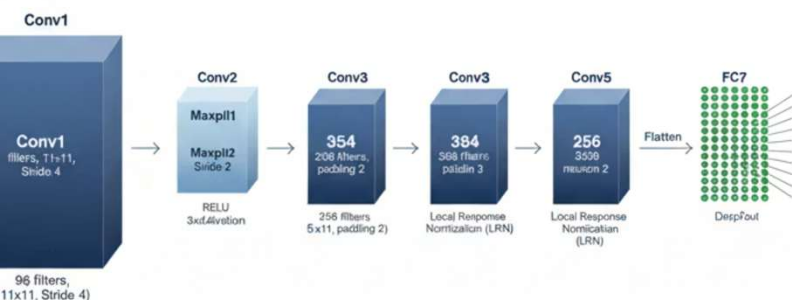
The Development Era (1970s–1980s)



This era laid the mathematical and algorithmic groundwork — texture analysis, color theory, and geometric transforms — that underpins every modern vision system.

AlexNet Architecture (2012)

Image Classification Model (ILSVRC-2012 Winner)



The Modern Era (1990s–2010s)

1990s

Face Recognition systems become viable. Eigenfaces and later Viola-Jones enable real-time detection

2000s

Real-time processing on commodity hardware. SIFT, HOG descriptors power robust feature matching

2012

AlexNet — Deep Learning revolutionizes CV, slashing ImageNet error rates by nearly 10%

2014+

YOLO, SSD, Faster R-CNN — real-time object detection at unprecedented accuracy

Recent Advances (2018–2024)



Vision Transformers (ViT)

Attention-based architectures applied to images, achieving state-of-the-art results across classification, detection, and segmentation benchmarks



SLAM + Deep Learning

Neural networks enhance simultaneous localization and mapping with semantic understanding and robustness to dynamic environments



3D Vision & Depth Estimation

Monocular depth estimation, NeRF, and 3D Gaussian Splatting reconstruct rich 3D scenes from standard cameras



Multi-Modal AI

Models like GPT-4V and CLIP combine visual and language understanding — enabling robots to follow natural language instructions



PART 4

Computer Vision in Robotics

Vision transforms a robot from a blind machine into an intelligent agent — able to perceive, reason about, and act upon the visual world around it.

Why Do Robots Need Vision?



Object Grasping

Detect, localize, and estimate the pose of objects before reaching out to grasp them reliably



Obstacle Avoidance

Perceive and react to unexpected obstacles in real time for safe autonomous movement



Localization (SLAM)

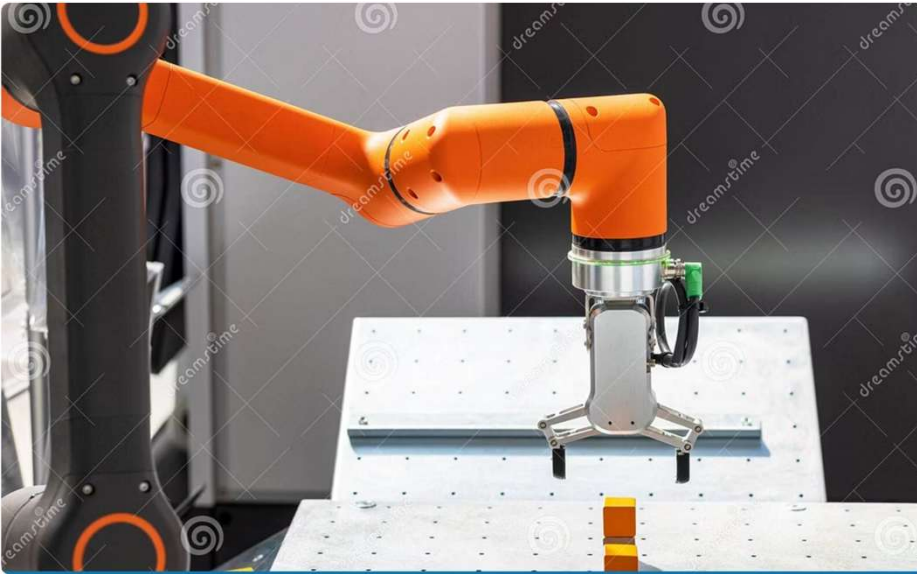
Build and maintain a map of the environment while simultaneously tracking the robot's position within it



Human Interaction

Recognize human faces, gestures, and intent to enable safe and natural collaboration in shared spaces

Vision Applications in Robotics



Pick & Place

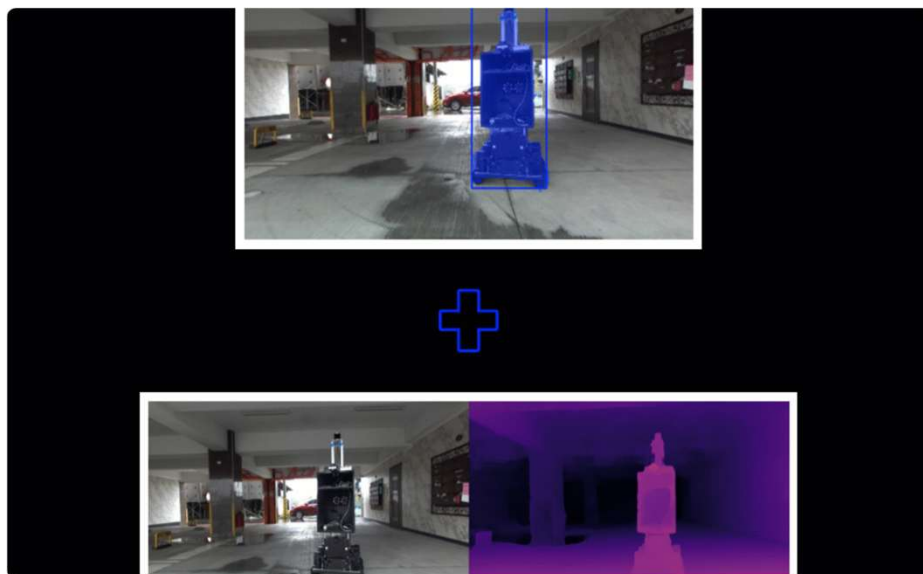
Vision systems identify objects scattered in a bin — detecting their position, orientation, and type — enabling robots to grasp items reliably from unstructured heaps. Used widely in e-commerce fulfillment and food processing.



Quality Control

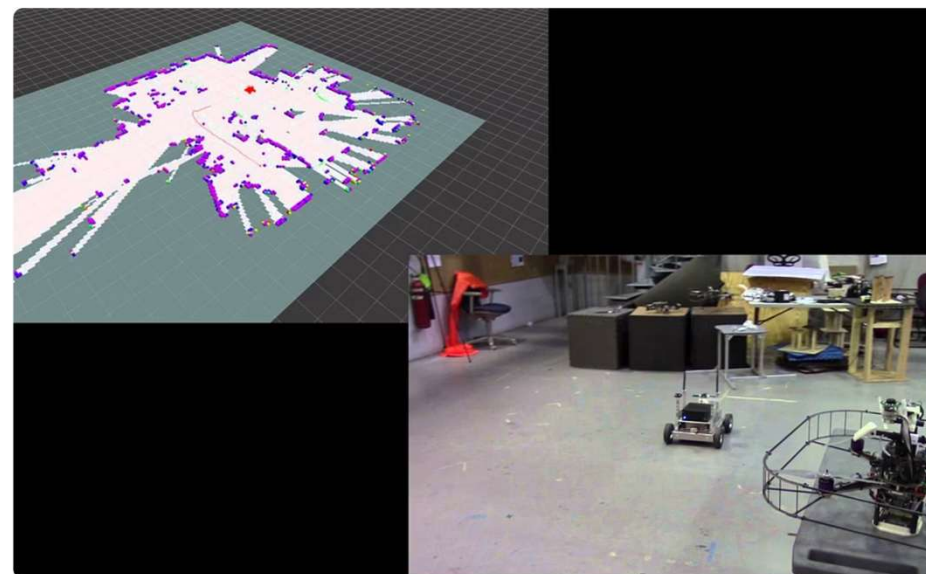
High-resolution cameras scan products on the assembly line at high speed, flagging defects, dimensional errors, and surface anomalies that would take human inspectors much longer to identify.

Vision Applications in Robotics



Visual Navigation

Robots use camera feeds and image processing to estimate motion, detect landmarks, and navigate safely through complex indoor and outdoor environments — even without GPS.



SLAM — Simultaneous Localization & Mapping

By continuously processing visual data, a robot builds a map of an unknown environment in real time while tracking its own position within that map — the foundation of all autonomous navigation.

Hand-Eye Calibration

The Core Challenge

Hand-eye calibration establishes the precise geometric relationship between the robot's camera (eye) and its end-effector (hand). Without it, a robot cannot reliably translate what it sees into where it should move.

The Pipeline

01

Perceive

Camera captures the scene and detects the target object's position

02

Transform

Compute the 6-DOF pose of the object in the robot's coordinate frame

03

Plan & Execute

Send trajectory commands to move the arm precisely to the target

Real-World Examples

Tesla Autopilot

Full self-driving system relying solely on cameras and AI — no LiDAR. Processes 8 camera streams simultaneously to navigate complex road environments.

Amazon Warehouse Robots

Thousands of Kiva/Proteus robots use computer vision to identify shelving units, navigate fulfillment centers, and handle package recognition.

Boston Dynamics

Atlas and Spot use stereo cameras and depth sensors to navigate unstructured environments, climb stairs, and manipulate objects in dynamic scenes.

Medical Robots

da Vinci surgical system uses 3D stereo cameras to give surgeons magnified, high-resolution views during minimally invasive procedures.



The Robotics + Vision Integration Framework



This closed-loop pipeline forms the backbone of every vision-guided robot system. Each stage depends on the previous one — a failure in perception cascades into incorrect action. Understanding each layer is a core learning objective of this course.

Course Structure

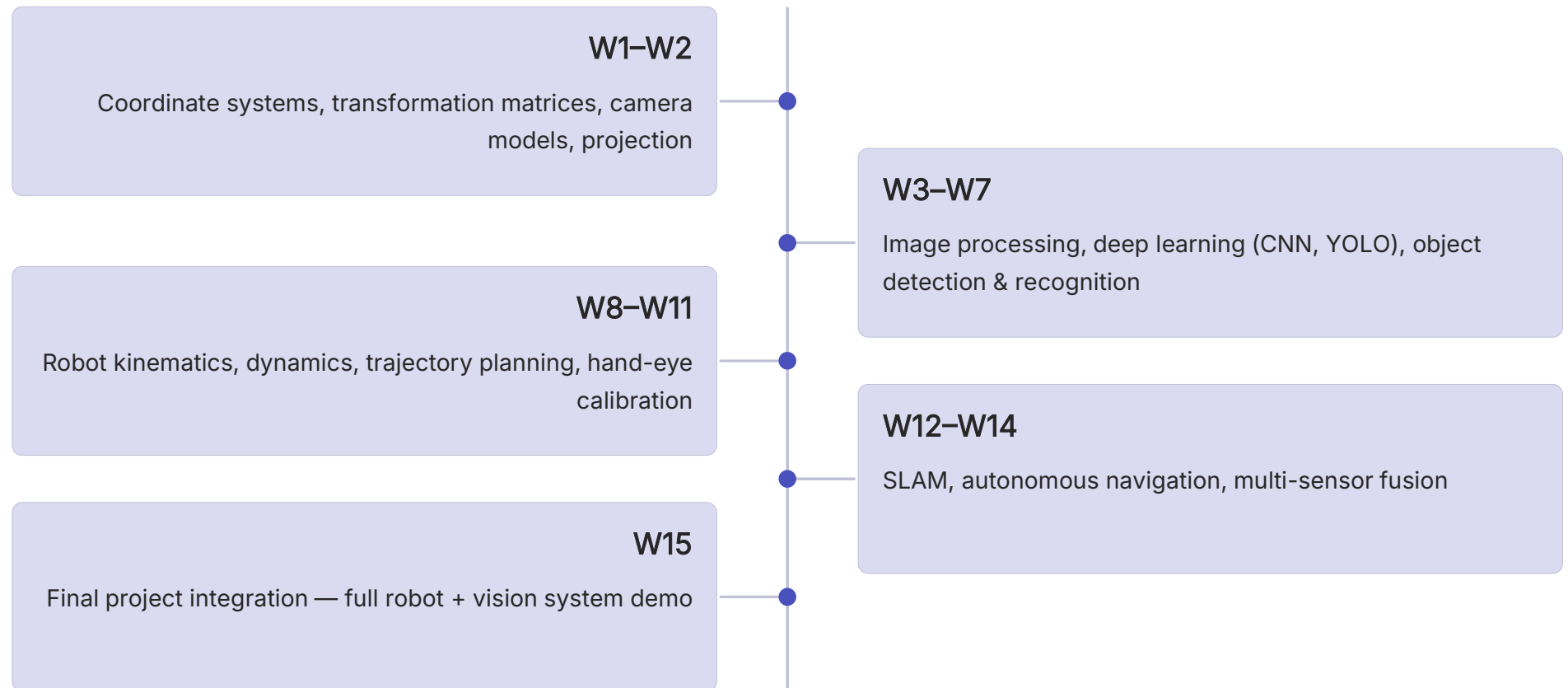
Weeks 1–7: Computer Vision Foundations

- Coordinate systems & transformation matrices
- Camera models and projection geometry
- Image processing fundamentals
- Deep learning for vision (CNNs, YOLO)
- Object recognition and tracking

Weeks 8–14: Robotics + Vision Integration

- Robot kinematics & dynamics
- Motion planning & trajectory control
- Hand-eye calibration
- SLAM and autonomous navigation
- Final project: full system integration

Learning Journey — Week by Week



Skills You Will Gain

Python + Toolbox

Fluency in Python for robotics with Peter Corke's Robotics Toolbox

Computer Vision

Classical algorithms and deep learning pipelines using OpenCV and PyTorch

Deep Learning

Training and deploying YOLO and CNN models for real-time perception tasks

Robot Kinematics

Forward & inverse kinematics, dynamics, and motion planning from first principles

SLAM & Navigation

Build maps, localize robots, and plan collision-free paths in unknown environments

Tools & Technologies



Python 3.x + Jupyter

The primary development environment — interactive notebooks for rapid prototyping and experimentation



Robotics Toolbox

Peter Corke's open-source library for kinematics, dynamics, trajectory planning, and robot simulation



OpenCV + NumPy

Industry-standard computer vision library for image processing, camera calibration, and feature detection



PyTorch / TensorFlow

Deep learning frameworks for training custom models and deploying pre-trained networks like YOLO



Gazebo / Webots

Full 3D robot simulators — prototype and validate your algorithms safely before running on real hardware

Real Hardware in This Course



Yahboom 6-DOF Arm

A desktop 6-joint manipulator — perfect for learning kinematics, pick-and-place, and hand-eye calibration experiments



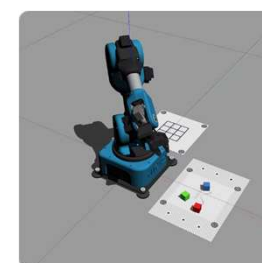
Jet Racer

NVIDIA Jetson-powered miniature self-driving car — used for lane following, obstacle avoidance, and vision-based control



Turtlebot 4

Full-featured ROS2 mobile robot with depth camera and LiDAR — ideal for SLAM and autonomous navigation projects



Simulator First

All tasks are prototyped in simulation before touching real hardware — a professional workflow that prevents costly mistakes

Assessment Structure

How You Are Evaluated

Assessment is designed to reward both **understanding** and **application**. Each component builds on the last, culminating in a full integrated robot-vision system.

20%

Case Study

Research and present a real-world computer vision application in robotics

40%

Weekly Exercises

Hands-on lab assignments completed each week — continuous practice

40%

Final Project

End-to-end robot + vision integration — the capstone of the course

Teamwork

The final project and many lab exercises are completed in teams of **2–3 students**. Effective collaboration is a core professional skill in robotics engineering.

Fair Division

Split tasks equitably
— hardware,
software, and
documentation each
need an owner

Regular Discussion

Meet weekly to sync
progress, resolve
blockers, and review
each other's code

Final Presentation

All team members
present results at
the end of the
course — everyone
must understand the
full system

Final Project Ideas



Person-Following Robot Arm

A robotic arm that detects and tracks a person's hand or body using pose estimation, continuously adjusting its position in real time



Color Line-Following Car

Jet Racer follows a colored path on the floor using camera-based color detection and PID control — classic autonomous navigation challenge



Automated Product Sorting

Classify objects by shape, color, or printed label using deep learning, then command the robot arm to sort them into the correct bin



SLAM + 3D Mapping

Use Turtlebot 4 with depth camera to build a 3D occupancy map of a room while localizing the robot — a full autonomous navigation showcase

Prerequisites

What You Need Coming In

This course assumes a solid foundation in programming and mathematics. You do not need prior robotics experience — but you must be comfortable with the following.



Python Basics

Functions, loops, NumPy arrays, and working with libraries



Linear Algebra

Matrices, vectors, dot products, and geometric transformations



Machine Learning Fundamentals

Core concepts from AI301 — loss functions, training, and model evaluation



Curiosity & Speed

A willingness to learn rapidly, experiment boldly, and ask questions

Key References

Peter Corke — Robotics, Vision & Control (Python Edition)

The primary textbook. Covers the entire curriculum using Python and the Robotics Toolbox. Available freely online at petercorke.com.

Richard Szeliski — Computer Vision: Algorithms & Applications (2nd Ed.)

Comprehensive reference for computer vision theory, from image formation to 3D reconstruction. Available free at szeliski.org.

Modern Robotics — Murray, Li & Sastry

Rigorous treatment of robot kinematics, dynamics, and control using modern geometric methods. Free PDF available online.

Online Resources

Coursera Robotics Specialization (Penn), YouTube: Cyrill Stachniss (SLAM), Andrej Karpathy (Deep Learning), and ROS documentation.

